

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279692

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

IMURA; Arashi et al.

VEHICLE DRIVE SYSTEM

Abstract

A vehicle drive system including a motor, a temperature sensor, and a resolver includes a motor housing accommodating the motor, the temperature sensor, and the resolver, a cover member covering the motor housing from one side in a motor axial direction, and temperature sensor and resolver harnesses connected to the temperature sensor and the resolver, respectively. The temperature sensor harness and the resolver harness are integrally provided inside the motor housing. The motor housing has a window through which a collective harness of the temperature sensor harness and the resolver harness passes. An external connector connectable to an external harness is attached to the cover member to pass through the cover member. The external connector is connected to a tip-end part of the collective harness.

Inventors: IMURA; Arashi (Aki-gun, JP), NAKANO; Atsushi (Aki-gun, JP)

Applicant: Mazda Motor Corporation (Aki-gun, JP)

Family ID: 94432874

Appl. No.: 19/040030

Filed: January 29, 2025

Foreign Application Priority Data

JP 2024-031405

Mar. 01, 2024

Publication Classification

Int. Cl.: H02K5/22 (20060101); H02K11/225 (20160101); H02K11/25 (20160101); H02K24/00 (20060101)

U.S. Cl.:

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a vehicle drive system which is installed in a vehicle.

BACKGROUND OF THE DISCLOSURE

[0002] For example, JP7083298B discloses a vehicle provided with a motor for driving wheels. This motor is an in-wheel motor built into a wheel hub, provided with a hollow shaft which rotates with a rotor, and a sensor which detects a rotational position of the rotor. A harness of the sensor is inserted through the hollow shaft and extends to the outside of the motor to be connected to a control board.

[0003] Meanwhile, an internal temperature state and a rotation state of the motor are used as part of information when controlling the motor. In order to detect the internal temperature of the motor as accurately as possible, it is necessary to place a temperature sensor inside a housing which accommodates the motor. Placing the temperature sensor inside the housing allows the sensor to be protected from an external environment, such as water and dust, which is thought to reduce issues like deterioration over time. Further, it is desired to place a resolver for detecting the rotation state of the motor inside the housing, in order to protect the resolver from the external environment.

[0004] When the temperature sensor and the resolver are placed inside the housing, harnesses extending from the temperature sensor and the resolver needs to be routed outside the housing. In this regard, JP7083298B routes the harness of the sensor through the hollow shaft to extend it outside. However, when the adoption of such a hollow shaft structure is difficult, how to route the harness outside the housing becomes an issue. In particular, since the temperature sensor and the resolver are positioned at different locations, it may be challenging to adopt a layout where both the harness of the temperature sensor and the harness of the resolver pass through the hollow shaft as disclosed in JP7083298B.

SUMMARY OF THE DISCLOSURE

[0005] The present disclosure is made in view of the above situations, and aims to output detection signals from a temperature sensor and a resolver to the outside of a housing, without adopting a hollow shaft structure, while facilitating a layout of a plurality of harnesses.

[0006] A vehicle drive system including a motor, a temperature sensor configured to detect a temperature state of the motor, and a resolver configured to detect a rotation state of the motor is provided, which includes a motor housing accommodating the motor, the temperature sensor, and the resolver, a cover member covering the motor housing from one side in an axial direction of a motor rotation shaft, a temperature sensor harness connected to the temperature sensor, and a resolver harness connected to the resolver. The temperature sensor harness and the resolver harness are integrally provided inside the motor housing. The motor housing includes a window through which a collective harness of the temperature sensor harness and the resolver harness passes. An external connector connectable to an external harness is attached to the cover member to pass through the cover member. The external connector is connected to a tip-end part of the collective harness.

[0007] The motor housing may include a housing-side vertical wall part extending in a radial direction of the motor rotation shaft, and a housing-side circumferential wall part projecting in the axial direction of the motor rotation shaft from a circumferential edge portion of the housing-side vertical wall part to extend annularly. The window may be formed in the housing-side vertical wall part.

[0008] The housing-side circumferential wall part may include an opening section openable and closable by a lid member. A coupling part may integrally provide the temperature sensor harness and the resolver harness inside the motor housing.

[0009] The resolver may be accommodated inside the motor housing on a cover member side. The temperature sensor may be accommodated inside the motor housing on an opposite side from the cover member.

[0010] The coupling part may be provided to an intermediate part of the temperature sensor harness.

[0011] The cover member may include a cover-side vertical wall part extending in the radial direction of the motor rotation shaft and a cover-side circumferential wall part extending in a circumferential direction of the motor rotation shaft. The external connector may be attached to the cover-side circumferential wall part to pass therethrough.

[0012] The cover member may accommodate a transmission thereinside.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIG. 1 is a front perspective view of a vehicle drive system.

[0014] FIG. 2 is a front view of the vehicle drive system.

[0015] FIG. 3 is a schematic cross-sectional view taken along a line III-III in FIG. 2.

[0016] FIG. 4 is an upper-left perspective view illustrating a disassembled state of a right cover member and a motor housing.

[0017] FIG. 5 is an upper-right perspective view illustrating the disassembled state of the right cover member and the motor housing.

[0018] FIG. 6 is a view indicating a state before the right cover member is assembled to the motor housing.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0019] Hereinafter, one embodiment of the present disclosure will be described in detail with reference to the accompanying drawings. Note that the following description of this embodiment is merely illustrative in nature and is not intended to limit the present disclosure, its applications, or uses.

[0020] FIGS. 1 and 2 illustrate a vehicle drive system 1 according to one embodiment of the present disclosure. The vehicle drive system 1 is mounted on an automobile (not illustrated) and propels driving wheels of the automobile. The automobile may be an electric vehicle which travels solely on power of a motor or a hybrid vehicle which uses both an internal combustion engine and the motor. The motor for generating the power to drive the drive wheels is a drive motor. The hybrid vehicle may be a plug-in hybrid vehicle which is chargeable from an external power source. In this embodiment, the explanation is given for a case where a vehicle on which the vehicle drive system 1 is installed is the hybrid vehicle. Therefore, as illustrated in FIG. 3, although the automobile on which the vehicle drive system 1 is installed is also equipped with a drive battery 100 which supplies power to a motor 2 and a controller 200 which controls the motor 2, since they are conventionally well-known, the detailed description thereof is omitted. The controller 200 includes, for example, an inverter circuit.

[0021] The vehicle drive system 1 is mounted, for example, in an engine compartment of the automobile. The engine compartment may be located in a front or rear part of the automobile. Front, rear, left, and right directions are defined as illustrated in each drawing. That is, with reference to the state where the vehicle drive system 1 is mounted in the engine compartment of the automobile, the front side of the vehicle is simply referred to as “front,” the rear side of the vehicle is simply referred to as “rear,” the left side of the vehicle is simply referred to as “left,” and the

right side of the vehicle is simply referred to as “right.” A left-and-right direction of the vehicle corresponds to a width direction of the vehicle. This definition of the directions is solely for the convenience in describing the embodiment and does not limit the present disclosure.

[0022] As partially illustrated by virtual lines in FIG. 2, an engine E is disposed inside the engine compartment, on the right side of the vehicle drive system 1. A crankshaft (not illustrated) of the engine E extends in the left-and-right direction, that is, the engine E is a so-called transverse engine. The engine E and the vehicle drive system 1 are arranged side by side in the left-and-right direction, the vehicle drive system 1 is disposed on the left side of the engine E, and a left part of the engine E is coupled to a right part of the vehicle drive system 1 by a fastening member (not illustrated). The power outputted from the crankshaft of the engine E is inputted to the vehicle drive system 1. Note that the left and right positional relationship may be reversed, and the arranged direction of the engine E and the vehicle drive system 1 is not particularly restricted. In the case of an electric vehicle, since the engine E is not installed, the engine compartment is referred to as a motor compartment.

[0023] FIG. 3 schematically illustrates a cross section of the vehicle drive system 1. As illustrated in FIG. 3, the vehicle drive system 1 includes the motor 2, a temperature sensor 3 which detects a temperature of the motor 2, and a resolver 4 which detects a rotation state of the motor 2. As illustrated in FIG. 4, the motor 2 includes a stator 2a and a rotor 2b (only an outer circumference thereof is partially illustrated by a virtual line). The core of a rotation shaft of the motor 2 (motor rotation shaft) is denoted by a reference character B in FIG. 2. A mounting direction of the vehicle drive system 1 in the vehicle is set so that a motor rotation shaft direction aligns with the left-and-right direction. Therefore, an axis core direction of the crankshaft of the engine E matches the motor rotation shaft direction.

[0024] The temperature sensor 3 is, for example, comprised of a thermistor and is disposed close to the stator 2a of the motor 2 in order to immediately detect a temperature change in the stator 2a. The temperature sensor 3 constantly outputs a detection signal. The resolver 4 is a rotational angle detection sensor which detects a rotational angle of the rotor 2b. The temperature sensor 3 and the resolver 4 may be of any commonly known type.

[0025] The vehicle drive system 1 also includes a motor housing 10, a right cover member 20, and a left cover member 30. The motor housing 10, the right cover member 20, and the left cover member 30 are each made of lightweight alloy, such as an aluminum alloy, providing high rigidity. Although in this embodiment the right and left cover members 20 and 30 are named this way since they are disposed on the respective right and left sides, without limiting to this, the right cover member 20 may be referred to as a front cover member and the left cover member 30 as a rear cover member. The right cover member 20 is a first cover member, and the left cover member 30 is a second cover member.

[0026] The motor housing 10, the right cover member 20, and the left cover member 30 form a single case A. The right cover member 20 is positioned on the right side of the case A to constitute a right part of the case A, and thus may also be referred to as a “right-side component.” The left cover member 30 is positioned on the left side of the case A to constitute a left part of the case A, and thus may also be referred to as a “left-side component.” The motor housing 10 is positioned between the right and left cover members 20 and 30 to constitute an intermediate part of the case A in the left-and-right direction, and thus may also be referred to as an “intermediate component.” The motor housing 10 is sandwiched in the left-and-right direction by the right and left cover members 20 and 30. The structure of the case A is not limited to the above structure and may include additional component(s).

[0027] The motor housing 10 forms a single accommodation space R (illustrated in FIG. 3). The motor 2, the temperature sensor 3, and the resolver 4 are accommodated inside the single accommodation space R. The motor housing 10 includes a housing-side vertical wall part 11 and a housing-side circumferential wall part 12, which are integrally molded. The housing-side vertical

wall part **11** is positioned at a left end part of the motor housing **10** and is an end wall part extending in a radial direction (up-and-down and front-and-rear directions) of the motor rotation shaft. The housing-side circumferential wall part **12** projects rightward along the motor rotation shaft direction from a circumferential edge portion of the housing-side vertical wall part **11** and extends continuously in a circumferential direction of the motor rotation shaft to form an annular wall part. In a tip end portion (right end portion) of the housing-side circumferential wall part **12**, a housing-side fastening portion **12a** protruding radially outward is formed. The housing-side fastening portion **12a** is joined to the right cover member **20** by a fastening member (not illustrated).

[0028] The motor **2** is positioned inside the housing-side circumferential wall part **12**. An output combining mechanism (not illustrated) for combining the output of the engine **E** with the output of the motor **2** (rotational force of the rotor **2b**) is provided inside the rotor **2b** (illustrated by virtual lines in FIG. **4**). The output combining mechanism is capable of inputting only the output of the engine **E**, only the output of the motor **2**, or a combination of the outputs of the engine **E** and the motor **2** at an arbitrary ratio, to a subsequent transmission **50** (illustrated in FIG. **3**). The structure of the output combining mechanism is not particularly limited and may apply a conventionally well-known mechanism used in hybrid vehicles. The control of the output combining mechanism may also use a conventionally well-known method.

[0029] The housing-side vertical wall part **11** is located in the left end part of the motor housing **10**, as the end wall part. However, since no end wall part is formed in the right end part of the motor housing **10**, the motor housing **10** has a structure where it has an opening formed in the right direction (one side in the motor rotation shaft direction). Since the accommodation space **R** opens in the right direction of the motor housing **10**, the opening of the motor housing **10** is covered by the right cover member **20** from the right side.

[0030] The right cover member **20** includes a right cover-side vertical wall part **21** and a right cover-side circumferential wall part **22**, which are integrally molded. The right cover-side vertical wall part **21** is an end wall part located in a left end part of the right cover member **20** and extending in the radial direction (up-and-down and front-and-rear directions) of the motor rotation shaft to cover the opening of the motor housing **10**. The housing-side fastening portion **12a** is joined to the right cover-side vertical wall part **21** to close the opening of the motor housing **10**.

[0031] The right cover-side circumferential wall part **22** projects from a circumferential edge portion of the right cover-side vertical wall part **21** in the motor rotation shaft direction and extends continuously in the circumferential direction of the motor rotation shaft to form an annular wall part. In a tip end portion (right end portion) of the right cover-side circumferential wall part **22**, a right cover-side fastening portion **22a** protruding radially outward is formed. The right cover-side fastening portion **22a** is connected to the engine **E** by a fastening member (not illustrated).

[0032] The left cover member **30** is attached to the motor housing **10** at the opposite side (left side) from the opening and includes a left cover-side vertical wall part **31** and a left cover-side circumferential wall part **32**, which are integrally molded. The left cover-side vertical wall part **31** is an end wall part located in a left end part of the left cover member **30** and extending in the radial direction (up-and-down and front-and-rear directions) of the motor rotation shaft.

[0033] The left cover-side circumferential wall part **32** projects from a circumferential edge portion of the left cover-side vertical wall part **31** in the motor rotation shaft direction and extends continuously in the circumferential direction of the motor rotation shaft to form an annular wall part. In a tip end portion (right end portion) of the left cover-side circumferential wall part **32**, a left cover-side fastening portion **32a** protruding radially outward is formed. The left cover-side fastening portion **32a** is coupled to the housing-side vertical wall part **11** of the motor housing **10** by a fastening member (not illustrated).

[0034] The transmission **50** is accommodated inside the left cover-side circumferential wall part **32**. The rotational force outputted from the output combining mechanism accommodated in the

motor housing **10** is inputted to the transmission **50** to be decelerated or accelerated and then be outputted. The output of the transmission **50** is transmitted to the drive wheels via a final gear and a drive shaft (not illustrated).

[0035] The temperature sensor **3** and the resolver **4** are spaced apart in the motor rotation shaft direction. The temperature sensor **3** is accommodated inside the motor housing **10** near the right cover-side vertical wall part **21** and, in this embodiment, is fixed by being fastened to the right cover-side vertical wall part **21**. The resolver **4** is accommodated inside the motor housing **10** on the opposite side from the right cover-side vertical wall part **21** and is fixed by being fastened to the housing-side vertical wall part **11**. Thus, the temperature sensor **3** is accommodated on the right side of the accommodation space R and the resolver **4** is accommodated on the left side of the accommodation space R.

[0036] The vehicle drive system **1** further includes a temperature sensor harness **3a** connected at its base end part (one end part) to the temperature sensor **3**, and a resolver harness **4a** connected at its base end part (one end part) to the resolver **4**. The temperature sensor harness **3a** contains a signal line which transmits the detection signal of the temperature sensor **3** to the controller **200** located outside the case A. The resolver harness **4a** contains a signal line which transmits the detection signal of the resolver **4** to the controller **200**.

[0037] The temperature sensor harness **3a** and the resolver harness **4a** are integrally provided inside the motor housing **10**. Specifically, a coupling part **60** which integrates the temperature sensor harness **3a** and the resolver harness **4a** is provided inside the motor housing **10**. The coupling part **60** is provided in an intermediate part of the temperature sensor harness **3a**. The temperature sensor harness **3a** is divided into a base-end harness **3b** located on the temperature sensor **3** side, and a tip-end harness **3c** located on a tip end side, from the coupling part **60**.

[0038] The coupling part **60** has a base-end connector **60a** connected to a tip end part of the base-end harness **3b**, and a tip-end connector **60b** connected to a base-end part of the tip-end harness **3c**. One of the base-end connector **60a** and the tip-end connector **60b** is a female connector and the other one is a male connector. By coupling the base-end connector **60a** to the tip-end connector **60b**, the base-end harness **3b** is conductively connected to the tip-end harness **3c**.

[0039] The vehicle drive system **1** further includes a collective harness **61** configured by collecting the temperature sensor harness **3a** and the resolver harness **4a** together. Specifically, the collective harness **61** is formed by collecting the tip-end harness **3c** of the temperature sensor harness **3a** and the resolver harness **4a**.

[0040] The motor housing **10** has on the left cover member **30** side a window **11a** through which the collective harness **61** passes. Specifically, the window **11a** is located in a front portion of the housing-side vertical wall part **11**, and is formed to extend through the housing-side vertical wall part **11** in the thickness direction (left-and-right direction). Forming the window **11a** allows the collective harness **61** (the tip-end harness **3c** and the resolver harness **4a**) to be routed from the inside of the motor housing **10** to the inside of the left cover member **30**. An inner surface of the window **11a** and the collective harness **61** can be sealed therebetween with a sealing material.

[0041] An external connector **62** connectable to an external harness **303** is attached to the left cover member **30** to pass therethrough. Specifically, the left cover-side circumferential wall part **32** has a connector attachment hole **32b** extending through the left cover-side circumferential wall part **32** in the thickness (radial) direction. The external connector **62** is fixed to the left cover-side circumferential wall part **32** in a state where it is inserted into the connector attachment hole **32b**.

[0042] A tip end part (other end part) of the collective harness **61** is connected to the external connector **62**. The external connector **62** is connectable to the external harness **303**. The side of the external connector **62** connected to the external harness **303** faces forward. This connection side may alternatively face upward, downward, or rearward.

[0043] The external harness **303** contains a signal line for transmitting the detection signals of the temperature sensor **3** and the resolver **4** to the controller **200**. A connector **304** connected to the

external connector **62** is provided in an end part of the external harness **303**. Since the connector **304** is connected to a front part of the external connector **62**, a rear part of the connector **304** is connected to the external connector **62**. Either of the external connector **62** and the second connector **304** may be a female or male connector.

[0044] For example, one of the external connector **62** and the connector **304** may have an insertion port to be a female connector, and the other one of the external connector **62** and the connector **304** may have an inserting part to be inserted into the insertion port, to be a male connector. If the external connector **62** has the insertion port, the arrangement of the external connector **62** may be determined so that the insertion port faces forward. In this case, by inserting the inserting part formed in the connector **304** into the insertion port of the external connector **62** from its front side, the connector **304** can be connected to the external connector **62**. Conversely, if the external connector **62** has the inserting part, the arrangement of the external connector **62** may be determined so that the inserting part projects forward.

[0045] The housing-side circumferential wall part **12** of the motor housing **10** has an opening section **12c** which is opened and closed by a lid member **70**. The opening section **12c** is a penetration hole through a front portion of the housing-side circumferential part **12** in the thickness direction (front-and-rear direction). The lid member **70** is disposed outside the housing-side circumferential wall part **12**, and the opening section **12c** can be closed from the outside of the housing-side circumferential wall part **12**. The lid member **70** is fixed to an outer surface of the housing-side circumferential wall part **12** by being fastened with a fastening member **72** (illustrated in FIG. 2, etc.).

[0046] As illustrated in FIGS. 4 and 6, before the right cover member **20** is attached to the motor housing **10**, the lid member **70** is not attached to the housing-side circumferential wall part **12** and the opening section **12** is uncovered. Further, the base-end connector **60a** and the tip-end connector **60b** are disassembled from each other and the tip-end connector **60b** is held to a circumferential edge portion of the opening section **12c** of the motor housing **10** in a state where it projects outside the motor housing **10** from the opening section **12c**.

[0047] After holding the tip-end connector **60b** to the circumferential edge portion of the opening section **12c** of the motor housing **10**, the right cover member **20** is attached to the motor housing **10**. Then, an operator is able to couple the tip-end connector **60b** to the base-end connector **60a** within the opening section **12c**. After coupling the tip-end connector **60b** to the base-end connector **60a**, the coupling part **60** is placed inside the motor housing **10**, and the lid member **70** is fixed to the outer surface of the housing-side circumferential wall part **12** by being fastened with the fastening member **72**. Thus, by forming the opening section **12c** which is openable/closable, the base-end connector **60a** and the tip-end connector **60b** can easily be coupled to each other.

Operation and Effect(s) of Embodiment

[0048] As described above, according to this embodiment, since the temperature sensor **3** for detecting the temperature state of the motor **2** is provided inside the motor housing **10**, the temperature state of the motor **2** may accurately be detected. Since the temperature sensor harness **3a**, which is connected to the temperature sensor **3**, is integrally provided inside the motor housing **10** with the resolver harness **4a**, which is connected to the resolver **4**, by the coupling part **60**, the temperature sensor harness **3a** and the resolver harness **4a** may be collected together to be the collective harness **61**. Further, the collective harness **61** is connected to the external connector **62** attached to the left cover member **30** in the state where it passes through the window **11a** of the motor housing **10**. Since the external harness **303** is connected to the external connector **62**, the detection signals of the temperature sensor **3** and the resolver **4** can be inputted into the controller **200** of the vehicle via the temperature sensor harness **3a**, the resolver harness **4a**, the collective harness **61**, and the external harness **303**.

[0049] Thus, by providing the external connector **62** in the left cover member **30**, it becomes possible to input the detection signals of the temperature sensor **3** and the resolver **4** to the

controller **200** without adopting the hollow shaft structure as in the conventional example. Further, since the temperature sensor harness **3a** and the resolver harness **4a** are collected together inside the motor housing **10**, it is sufficient to only have the collective harness **61** pass through the window **11a**, and it is sufficient to only have the collective harness **61** connect to the external connector **62**. Therefore, when the temperature sensor harness **3a** and the resolver harness **4a** are provided, it becomes easier to facilitate a layout.

[0050] The above embodiment is merely illustrative in every aspect and must not be interpreted restrictively. Moreover, any modifications and/or changes within the scope of equivalency of the appended claims are within the scope of the present disclosure.

Industrial Applicability

[0051] As described above, the vehicle drive system according to the present disclosure can be used in hybrid vehicles or electric vehicles, for example.

[0052] It should be understood that the embodiments herein are illustrative and not restrictive, since the scope of the invention is defined by the appended claims rather than by the description preceding them, and all changes that fall within metes and bounds of the claims, or equivalence of such metes and bounds thereof, are therefore intended to be embraced by the claims.

DESCRIPTION OF REFERENCE CHARACTERS

[0053] **1** Vehicle Drive System [0054] **2** Motor [0055] **3** Temperature Sensor [0056] **3a** Temperature Sensor Harness [0057] **4** Resolver [0058] **4a** Resolver Harness [0059] **10** Motor Housing [0060] **11** Housing-side Vertical Wall Part [0061] **11a** Window [0062] **12c** Opening Section [0063] **60** Coupling Part [0064] **62** External Connector [0065] **70** Lid Member [0066] **303** External Harness

Claims

1. A vehicle drive system including a motor, a temperature sensor configured to detect a temperature state of the motor, and a resolver configured to detect a rotation state of the motor, further comprising: a motor housing accommodating the motor, the temperature sensor, and the resolver; a cover member covering the motor housing from one side in an axial direction of a motor rotation shaft; a temperature sensor harness connected to the temperature sensor; and a resolver harness connected to the resolver, wherein the temperature sensor harness and the resolver harness are integrally provided inside the motor housing; wherein the motor housing includes a window through which a collective harness of the temperature sensor harness and the resolver harness passes, wherein an external connector connectable to an external harness is attached to the cover member to pass through the cover member, and wherein the external connector is connected to a tip-end part of the collective harness.
2. The vehicle drive system of claim 1, wherein the motor housing includes a housing-side vertical wall part extending in a radial direction of the motor rotation shaft, and a housing-side circumferential wall part projecting in the axial direction of the motor rotation shaft from a circumferential edge portion of the housing-side vertical wall part to extend annularly, and wherein the window is formed in the housing-side vertical wall part.
3. The vehicle drive system of claim 2, wherein the housing-side circumferential wall part includes an opening section openable and closable by a lid member, and wherein a coupling part integrally provides the temperature sensor harness and the resolver harness inside the motor housing.
4. The vehicle drive system of claim 3, wherein the resolver is accommodated inside the motor housing on a cover member side, and wherein the temperature sensor is accommodated inside the motor housing on an opposite side from the cover member.
5. The vehicle drive system of claim 4, wherein the coupling part is provided to an intermediate part of the temperature sensor harness.
6. The vehicle drive system of claim 1, wherein the resolver is accommodated inside the motor

housing on a cover member side, and wherein the temperature sensor is accommodated inside the motor housing on an opposite side from the cover member.

7. The vehicle drive system of claim 6, wherein the coupling part is provided to an intermediate part of the temperature sensor harness.

8. The vehicle drive system of claim 1, wherein the cover member includes a cover-side vertical wall part extending in a radial direction of the motor rotation shaft and a cover-side circumferential wall part extending in a circumferential direction of the motor rotation shaft, and wherein the external connector is attached to the cover-side circumferential wall part to pass therethrough.

9. The vehicle drive system of claim 8, wherein the cover member accommodates a transmission thereinside.

10. The vehicle drive system of claim 2, wherein the cover member includes a cover-side vertical wall part extending in the radial direction of the motor rotation shaft and a cover-side circumferential wall part extending in a circumferential direction of the motor rotation shaft, and wherein the external connector is attached to the cover-side circumferential wall part to pass therethrough.

11. The vehicle drive system of claim 10, wherein the cover member accommodates a transmission thereinside.
